Multilingual and Cross-lingual Fact-Checked Claim Retrieval

A Learning-to-Rank Approach for SemEval-2025 Task 7

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Problem Statement

The rapid spread of disinformation across multiple languages presents a major challenge to conventional fact-checking workflows. SemEval-2025 Task 7—Multilingual and Cross-lingual Fact-Checked Claim Retrieval (Peng et al. 2025)—aims to address this by automating the retrieval of previously verified claims across diverse linguistic contexts. The dataset includes fact-checked claims and social media content in eight languages (French, Spanish, English, Portuguese, Thai, German, Malay, and Arabic), with additional evaluation in two unseen languages (Polish and Turkish) to test generalization.

To tackle this challenge, our approach focuses on both multilingual and cross-lingual retrieval using lightweight transformer models that are scalable and resource-efficient. By combining rich semantic embeddings with techniques like layer freezing and gradient checkpointing, we strike a balance between performance and computational efficiency. The result is a practical, adaptable system for combating global misinformation through accurate and scalable claim retrieval.

Background

SemEval-2025 Task 7 focuses on automating the retrieval of verified fact-checks for social media posts, especially across different languages—a task that is time-consuming if done manually. It includes two key subtasks:

Multilingual Retrieval: Post and fact-check are in the same language.

Cross-lingual Retrieval: Post and fact-check are in different languages, requiring robust cross-language matching techniques.

Model Architecture

Our system uses a **bi-encoder retrieval architecture** (Reimers and Gurevych 2019), where separate encoders process social media posts and fact-checked claims. Each encoder is initialized with a **pretrained transformer**, producing contextual embeddings for a given input sequence $X = (x_1, x_2, ..., x_n)$, yielding hidden states:

$$H = M(X) \in \mathbb{R}^{n \times d}$$

To obtain fixed-size sentence embeddings, we explore multiple **pooling strategies**. **Mean Pooling** computes an average over token embeddings weighted by attention masks:

$$\text{MeanPooling}(H, A) = \frac{\sum_i H_i \cdot A_i}{\sum_i A_i + \epsilon}$$

We also use an **attention-based BiLSTM pooling** strategy:

$$L = \text{BiLSTM}(H), \quad \alpha = \text{softmax}(W_a L), \quad S = \sum_i \alpha_i L_i$$

The similarity between post and claim embeddings q and c is measured using **temperature-scaled cosine similarity**:

$$S(q, c) = \frac{\cos(q, c)}{T}, \quad T = 0.05$$

We optimize training with **Multiple Negative Ranking (MNR) loss** (Henderson et al. 2017):

$$\mathcal{L}_{\text{MNR}} = -\log \frac{\exp(S_{ii})}{\sum_j \exp(S_{ij})}$$

And a **symmetric cross-entropy formulation**:

$$\mathcal{L} = \frac{1}{2}\text{CE}(S, y) + \text{CE}(S^\top, y)$$

To enable scalable training on modest hardware, we apply **layer freezing** and **gradient checkpointing**, making the model efficient and effective for multilingual claim retrieval.

Training Strategy

Model	Cross-Lingual (avg)	Multilingual (avg)	eng	fra	deu	por	spa	tha	msa	ara	tur	pol
multilingual-e5-large-instruct (Best Fold)	0.7685	0.9091	0.864	0.932	0.904	0.85	0.924	0.967	1.0	0.942	0.864	0.848
stella-en-400M-v5 (Best Fold)	0.76	0.9095	0.852	0.913	0.902	0.84	0.917	0.978	1.0	0.931	0.848	0.874
mxbai-embed-large-v1 (Best Fold)	0.7275	0.8972	0.844	0.92	0.884	0.828	0.902	0.978	0.989	0.932	0.844	0.852
multilingual-e5-large-instruct (Ensemble)	0.7975	0.9232	0.882	0.944	0.926	0.866	0.942	0.995	0.989	0.940	0.884	0.864
stella-en-400M-v5 (Ensemble)	0.7638	0.9140	0.876	0.936	0.906	0.870	0.946	0.978	1.000	0.932	0.844	0.852
mxbai-embed-large-v1 (Ensemble)	0.7712	0.9146	0.852	0.934	0.892	0.868	0.930	0.989	0.978	0.956	0.886	0.860

Posts and fact-checked claims are encoded using weight-sharing bi-encoders. Sentence embeddings are generated via either **Mean Pooling** or **BiLSTM-based attention** (Hochreiter and Schmidhuber 1997). We compute similarities using **temperature-scaled cosine similarity** and optimize the model with the contrastive loss functions described earlier.

We initialized encoders with three efficient pre-trained models, each having fewer than 500M parameters to ensure compatibility with free GPU resources:

NovaSearch/stella-en-400M-v5(Zhang et al. 2024): 435M parameters, ranked 41st on MTEB; selected for its balance between scalability and accuracy.

intfloat/multilingual-e5-large-instruct(Wang et al. 2024): 560M parameters (we apply layer freezing to stay under compute limits), 24-layer XLM-RoBERTa model supporting 100+ languages; ranked 22nd on MTEB.

mixedbread-ai/mxbai-embed-large-v1(Lee et al. 2024): 335M parameters, ranked 49th on MTEB; selected for its strong English representation.

All models were fine-tuned using **gradient checkpointing** and **selective layer freezing** to optimize the trade-off between training cost and retrieval accuracy. We used English-translated versions of the dataset to reduce overfitting and boost generalization, particularly in low-resource cross-lingual setups.

To improve robustness, we applied a **5-fold ensembling** strategy. This approach yielded consistent performance gains across all models. For example, the multilingual-e5-large-instruct model improved from 0.7685 to 0.7975 in cross-lingual Success@10, and from 0.9091 to 0.9232 in multilingual Success@10.

We also evaluated other top-ranking MTEB models such as `jina-embeddings-v3` and `KaLM-embedding-multilingual-mini-v1`, but their average scores (58.37 and 57.05 respectively) were outperformed by our primary choices, particularly multilingual-e5-large-instruct, which achieved an average of 63.23.

For **multilingual training**, we trained the bi-encoder (Reimers and Gurevych 2019) on the full dataset across all source languages, benefiting especially low-resource languages such as German, French, Malay, and Thai. Multiple training epochs allowed the model to effectively learn semantic and syntactic structures without using synthetic data.

In the **cross-lingual setting**, we reused the same bi-encoder models but trained them on English-translated text. To encourage generalization, we applied a **5-fold data split**, training on different distributions to reduce overfitting. We found that **ensembling English-trained models** was more effective than synthetic augmentation for cross-lingual generalization.

This ensemble strategy led to measurable improvements: for instance, multilingual-e5-large-instruct improved from 0.7685 to 0.7975 in cross-lingual Success@10 and from 0.9091 to 0.9232 in multilingual retrieval. We selected 30% of negative examples randomly during training (with a fixed seed of 42) to construct challenging batches. All models remained under 500M parameters and were trained using **public GPU resources**

Parameter	Value
Batch Size	24
Number of Epochs	20
Warmup Steps	400
Mixed Precision Training	`float16`

Table 3: MultiLingual Training Configuration

Parameter	Value
Batch Size	36
Number of Epochs	10
Warmup Steps	500
Mixed Precision Training	`float16`

Table 4: Cross Lingual Training Configuration

Results

Our multilingual model ranked **10th overall**, with an average Success@10 (S@10) score of **0.9232**. While it delivers competitive performance across most languages, it trails top-ranked systems in consistency.

Metric	Score	Rank
S@10 (avg)	0.923178	10.0
S@10 (eng)	0.882	7.0
S@10 (fra)	0.944	5.0
S@10 (deu)	0.926	6.0
S@10 (por)	0.866	9.0
S@10 (spa)	0.942	7.0
S@10 (tha)	0.994536	4.0
S@10 (msa)	0.989247	8.0
S@10 (ara)	0.94	9.0
S@10 (tur)	0.884	10.0
S@10 (pol)	0.864	10.0

Best-performing languages include Thai (S@10 = 0.9945, Rank = 4th) and French (S@10 = 0.944, Rank = 5th), where the model consistently ranks in the upper half. Performance in English is moderate (Rank = 7th), indicating that the model is not overly tuned to high-resource languages but instead maintains a cross-lingual balance.

However, **weaker performance** is observed in Polish (S@10 = 0.864, Rank = 10th) and Turkish (S@10 = 0.884, Rank = 10th), highlighting challenges in generalizing to morphologically rich or unseen languages. The wide range in performance (S@10 from 0.864 to 0.9945) suggests strong **generalization** in under-represented languages like Thai and Malay, while the drop in some mid-resource languages signals risks of **domain overfitting** or dataset imbalance.

Our cross-lingual system ranks **5th overall**, with an average S@10 score of **0.7975**, placing it firmly in the top half of systems and validating its effectiveness in cross-language retrieval.

Rank	Team Name	S@10 (avg)
1	PINGAN AI	0.85875
2	PALI	0.82675
3	Sherlock	0.8245
4	TIFIN India	0.81025
5	**fact check AI**	**0.7975**

Conclusion

We assessed the performance of various multilingual embedding models for fact-checked claim retrieval across both multilingual and cross-lingual settings. The results demonstrate that model effectiveness varies significantly by language, particularly in low-resource or morphologically rich cases such as Polish and Turkish. The **multilingual-e5-large-instruct** model consistently performed best, showing strong results in both single-language and cross-language retrieval tasks. In contrast, **stella-en-400M-v5** and **mxbai-embed-large-v1** were less impactful in isolated language settings.

While ensembling and English-translated data improved generalization, the system still exhibited vulnerabilities—particularly when semantically shallow cues (like shared named entities) led to fact-check mismatches. These observations highlight the importance of deeper semantic modeling and language-aware adaptation to build robust multilingual fact-checking systems.

Citation

Data expansion: Incorporate more data for under-represented and morphologically rich languages.

Domain adaptation: Fine-tune on domain-specific datasets to reduce factual mismatches.

Error mitigation: Address risks of false positives by improving sensitivity to deeper semantic and contextual distinctions.